Hence, $2\tan^{-1}x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right),\ x \geq 0$.

Similarly, other result is proved.

Property-VIII

(i) $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\sin^{-1}x$ if $|x| \leq \frac{1}{\sqrt{2}}$ or $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$.

(ii) $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\cos^{-1}x$ if $\frac{1}{\sqrt{2}} \leq x \leq 1$.

Proof

(i) Let $x = \sin\theta$.

Now, $2x\sqrt{1-x^2} = 2\sin\theta\cos\theta = \sin 2\theta$

Thus, $2\theta = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$. Hence, $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\sin^{-1}x$.

(ii) Let $x = \cos\theta$.

Now, $2x\sqrt{1-x^2} = 2\cos\theta\sin\theta = \sin 2\theta$, which gives

$2\theta = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$. Hence, $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\cos^{-1}x$.

Property-IX

(i) $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$, if $0 \leq x \leq 1$. (ii) $\sin^{-1}x = -\cos^{-1}\sqrt{1-x^2}$, if $-1 \leq x < 0$.

(iii) $\sin^{-1}x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$, if $-1 < x < 1$. (iv) $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2}$, if $0 \leq x \leq 1$.

(v) $\cos^{-1}x = \pi - \sin^{-1}\sqrt{1-x^2}$, if $-1 \leq x < 0$. (vi) $\tan^{-1}x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$, if $x > 0$.

Proof

(i) Let $\sin^{-1}x = \theta$. Then, $\sin\theta = x$. Since $0 \leq x \leq 1$, we get $0 \leq \theta \leq \frac{\pi}{2}$.

$\cos\theta = \sqrt{1-x^2}$ or $\cos^{-1}\sqrt{1-x^2} = \theta = \sin^{-1}x$.

Thus, $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$, $0 \leq x \leq 1$.

(ii) Suppose that $-1 \leq x \leq 0$ and $\sin^{-1}x = \theta$. Then $-\frac{\pi}{2} \leq \theta < 0$.

So, $\sin\theta = x$ and $\cos(-\theta) = \sqrt{1-x^2}$ (since $\cos\theta > 0$)

Thus, $\cos^{-1}\sqrt{1-x^2} = -\theta = -\sin^{-1}x$. Hence, $\sin^{-1}x = -\cos^{-1}\sqrt{1-x^2}$.

Similarly, other results are proved.

Property-X

(i) $3\sin^{-1}x = \sin^{-1}(3x-4x^3)$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$. (ii) $3\cos^{-1}x = \cos^{-1}(4x^3-3x)$, $x \in \left[\frac{1}{2}, 1\right]$.

Chapter 4 Inverse Trigonometry.indd 159 31-01-2020 17:50:15